



Academy of
Engineering

(An Autonomous Institute affiliated to Savitribai Phule Pune University)

Bloom's Algorithm

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Code:

```
#include <bits/stdc++.h>

#define ll long long using
namespace std;

// hash 1 int h1(string s, int arrSize) {
ll hash = 0;   for (int i = 0; i < s.size();
i++) {       hash = (hash + (int)s[i]) %
arrSize;
    }
    return hash;
}
```

```
// hash 2 int h2(string s, int
arrSize) {    ll hash = 1;
ll power = 1;

    for (int i = 0; i < s.size(); i++) {        hash = (hash +
power * s[i]) % arrSize;        power = (power * 19) %
arrSize; // replaces pow(19, i)
    }
    return hash;
}
```

```
// hash 3 int h3(string s, int arrSize) {
ll hash = 7;    for (int i = 0; i < s.size();
i++) {        hash = (hash * 31 + s[i]) %
arrSize;
    }
    return hash;
}
```

```
// hash 4 int h4(string s, int
arrSize) {    ll hash = 3;    ll
power = 1;    int p = 7;

    for (int i = 0; i < s.size(); i++) {        hash = (hash + hash
* 7 + s[i] * power) % arrSize;        power = (power * p) %
arrSize; // replaces pow(p, i)
```

```

    }    return
hash;
}

// lookup bool lookup(bool* bitarray, int
arrSize, string s) {    int a = h1(s, arrSize);    int
b = h2(s, arrSize);    int c = h3(s, arrSize);    int
d = h4(s, arrSize);

    return (bitarray[a] && bitarray[b] && bitarray[c] && bitarray[d]);
}

// insert void insert(bool* bitarray, int arrSize,
string s) {    if (lookup(bitarray, arrSize, s)) {
cout << s << " is probably already present\n";

    } else {        int a =
h1(s, arrSize);        int b =
h2(s, arrSize);        int c =
h3(s, arrSize);        int d =
h4(s, arrSize);
bitarray[a] = true;
bitarray[b] = true;
bitarray[c] = true;
bitarray[d] = true;

        cout << s << " inserted\n";
    }
}

```

```
}
```

```
int main() {    int arrSize =  
100;    bool bitarray[100] =  
{false};
```

```
    int choice;  
    string input;
```

```
    while (true) {        cout << "\n--- Bloom  
Filter Menu ---\n";        cout << "1. Insert  
string\n";        cout << "2. Lookup string\n";  
        cout << "3. Exit\n";        cout << "Enter  
choice: ";        cin >> choice;
```

```
        if (choice == 1) {            cout <<  
"Enter string to insert: ";            cin >>  
input;
```

```
        insert(bitarray, arrSize, input);
```

```
    }
```

```
    else if (choice == 2) {        cout << "Enter  
string to lookup: ";        cin >> input;        if  
(lookup(bitarray, arrSize, input))            cout <<  
input << " is probably present\n";
```

```
        else
```

```
            cout << input << " is definitely NOT present\n";
```

```
    }  
    else if (choice == 3) {  
cout << "Exiting...\n";  
break;    }    else {    cout  
<< "Invalid choice!\n";  
    }  
    }  
  
    return 0;  
}
```

Output:

```
--- Bloom Filter Menu ---
1. Insert string
2. Lookup string
3. Exit
Enter choice: 1
Enter string to insert: samyak
samyak inserted

--- Bloom Filter Menu ---
1. Insert string
2. Lookup string
3. Exit
Enter choice: 1
Enter string to insert: narayan
narayan inserted

--- Bloom Filter Menu ---
1. Insert string
2. Lookup string
3. Exit
Enter choice: 2
Enter string to lookup: samyak
samyak is probably present
```

```
--- Bloom Filter Menu ---
1. Insert string
2. Lookup string
3. Exit
Enter choice: 2
Enter string to lookup: aryan
aryan is definitely NOT present

--- Bloom Filter Menu ---
1. Insert string
2. Lookup string
3. Exit
Enter choice: █
```